The Art of Convincing: Can AI Out-Persuade Humans?

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CHAPTER 1: INTRODUCTION

1.1 Background

Persuasion is highlighted in relation to health, purchases, or policy-making in general it is not a slight matter, and is generally aimed at transforming the opinions of people, rather than merely pouring information on them. With big language models providing advice, explanations and arguments now, the question that has become an empirical question of its own is whether the arguments provided by AI can persuade people just as successfully as human arguments, and in what circumstances (Fischetti, 2025). Anthropic's Persuasion Dataset was deployed to test precisely this, with short claims of policy relevance being paired with arguments generated by a human or by a model, and observing the variation in agreement to an argument after reading arguments perceived by a reader. The experiment has the participants rate their position on a claim first, read an argument, and rate afterwards. The data consists as the starting and finishing ratings and metadata on the author of the argument, such as human or a particular model, and the type of prompt that was taken to produce the arguments of the model. Their breakdown reveals that the persuasiveness is projected to become better with the more recent the model generation and that Claude 3 Opus generated the arguments that were statistically identical to human ones in their experiment (Barale, 2021). This preconditions the analysis of what features of claims and arguments cause persuasion and whether some of the prompting strategies equally enhance persuasion.

1.2 Aim and Objectives

The study aims to understand whether an AI-based argument can be as persuasive and effective as an argument a human should write and to identify the elements of an argument and prompt that drive those changes in attitude. The objectives of the study are:

- To measure persuasion changes by calculating the shift from rating_initial to rating_final across argument sources and prompt_type conditions.
- To identify what drives persuasion by testing which claim and argument features (textual and contextual) predict higher final ratings and larger rating shifts.
- To compare prompting strategies to determine which prompt types produce the most persuasive AI arguments relative to human arguments.

1.3 Research Question

- What are the characteristics of claims and arguments that influence their initial and final persuasiveness ratings, and how do different prompt types and sources (e.g., AI models like Claude) affect these metrics?

1.4 Rationale of the Study

The study is justified on both scientifically and on a social front. Scientifically, the art of persuasion is a very complex behavioural product: the success or failure of an argument is predetermined by the controversialness of a proposition at the beginning of it, the priori attitude of the reader, and message rhetoric. The Anthropic data is convenient in the sense that it organises the idea of persuasion through post-and-pre stance ratings and the correlations between such results directly with the text of the argument, its source and the prompt that was applied to come up with the sentiment (Soliman, 2024). It to go beyond them not being sure who is right and discuss what goes in to an argument working: clarity, laying evidence in perspective, the use of morality, and the combination of emotion and rationality.

The power of scaling the persuasive messages has both advantages and disadvantages to the society. AI could be used to persuade people about new policies, make more healthy choices, or enhance education; in other words, persuasive AI. Nonetheless, it may also be used as a weapon in case systems are adjusted to favour and not the truth(Dehnert and Mongeau, 2022). Anthropic went so far as to include an inaccurate prompt that allows fabricated facts and sources which highlights the importance of prompt design in terms of performance and safety. Through cross-comparison of

prompt types and sources within a similar measurement framework, this project will be able to identify which aspects of AI persuasion resemble human persuasion, where it is different, and what aspects of AI persuasion respond strongest to which levers, such as prompting strategies. That fact proves handy when the researchers create assessment approaches and when practitioners create less harmful, more responsible AI which communicates with human thoughts and choices (Luciano, 2024).

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Chapter 2 begins by situating the current research within a broader academic framework in which persuasion is understood as the deliberate use of communicative information to affect an interlocutor's attitudes, beliefs, or behaviours. Persuasion is a phenomenon that social psychology and communication scholars have long focused on; it is regularly studied using theoretical frameworks, such as dual-process models that distinguish between deliberative and heuristic message processing, or through empirical research that disaggregates source, content, and audience influences on attitude change. With the introduction of large language models and conversational AI, this field has begun to incorporate the nature of how intelligent systems can produce, alter, and transmit messages designed to manipulate human cognition and behaviour. It has also demonstrated the incorporation of empirical research that can measure any changes in attitude pre- and post-exposure to such AI-generated arguments. The literature review included in this paper is limited to peer-reviewed articles that use stringent empirical methods to quantify the impact of persuasion, e.g., comparing AI-mediated versus human-mediated persuasion through controlled experiments, and theoretical papers that offer an analytical framework for AI-mediated persuasion. These selection criteria will ensure that the review focuses on studies that measure persuasive efficacy and inform the strategy for studying AI in persuasive environments, thereby providing a basis for further comparison of AI and human-generated arguments in the given project.

2.2 Review of Relevant Studies

2.2.1 Paper 1: Lin et al. (2025) — Persuading voters using human–AI dialogues (Nature)

Lin et al. (2025) provide one of the first empirical measurements, to date, of the breadth of AI systems' ability to modify individuals' substantive political views through conversation. They have pre-registered for experiments in the 2024 United States presidential election and the 2025 general elections in Canada and Poland. Subjects were randomly assigned to a group that interacted with a large-scale language model, were instructed to support a particular leading candidate, and had their partisan choice evaluated before and after the interaction. In these different national settings, the researchers have found statistically significant changes in voter preferences that are more often reported by the traditional campaign media, including from television advertisements, and have found explicit persuasive impacts on non-electoral topics, such as enhanced support for a ballot measure in Massachusetts. The overall analysis reveals that the AI models focused first on presenting facts and related evidence and on persuading participants, rather than on advanced psychological manipulation. However, the validity of the claims varied, with right-leaning positions containing more deceptive content.

The research is also consequential to the current project, as it shows, on a large scale, that AI-generated dialogues can trigger attitudinal change in a setting that realistically mimics real-life decision-making. The comparison between AI and human persuasive effects is supported by pre-registered, randomised experimental designs and quantified changes in preferences. In addition, it underscores a salient interaction between content accuracy and persuasive effectiveness, consistent with the thesis's overall purpose: to examine which textual and contextual properties of arguments predict both the extent and direction of persuasion. Although the results are very educative, the narrow area of political persuasion that the study was directed makes it difficult to generalise to other areas including health or policy claims out of the electoral setting and the study focuses in general upon the overall opinion change and not directly trying to break down the textual features or rhetorical constructs that form the background of the success of persuasion. In its turn, it means that although Lin et al. (2025) give a compelling argument about how attitude can be shifted through AI, the further analysis should complement their findings with the studies that can isolate the definite features of the arguments that cause such shifting and investigate how they depend on the type of prompt and the type of model used.

2.2.2 Paper 2: Dehnert and Mongeau (2022) — Persuasion in the age of artificial intelligence (Human Communication Research)

Dehnert and Mongeau's (2022) article in Human Communication Research does not attempt to quantify persuasion's effects empirically, but rather presents a conceptual model that reorganises the definition of persuasion in the age of artificial intelligence. The authors conceive of AI-based persuasion as a symbolic process in which a communicative AI object creates, adds to, or alters a message aimed at shaping the attitudes or behaviours of a human recipient. This definition extends classical models of persuasion by treating artificial systems not just as tools but as free communicative agents with an intentional sense of creating a message structure. Their synthesis of existing communication paradigms, including the Computers Are Social Actors (CASA) approach, and the use of other theoretical perspectives, including the Modality, Agency, Interactivity, and Navigability (MAIN) model and the heuristic-systematic models of persuasion. Combined, these views express how the different embodiment of AI, between the thin systems that possess few social cues and the thick ones that have more immersive interactive presence, can influence human interpretation of messages in various ways.

Dehnert and Mongeau (2022) make a direct contribution to the present project by establishing a theoretical framework for understanding AI-mediated persuasion beyond mere attitude-change indicators. Their work makes it clear that human beings might react differently when exposed to AI-generated arguments than to human-generated arguments, since it highlights how source cues and communicative context can affect message processing. The fact that they focus specifically on social and machine cues promotes studying the features of texts and contexts in arguments, which aligns with the project's purpose of determining factors of persuasion. Compared with empirical studies that report outcome data, this article combines existing theories to infer conceptual differences that future research must take into account; therefore, its major contribution is clarifying how AI agents could persuade, but not quantifying the magnitude of that persuasion.

One of the strengths of the work is that it combines traditional theories of communication with the new reality of human-AI interaction, offering a subtle way to examine AI as a communicative source. Such theoretical richness allows empirical findings in persuasion research to be interpreted more sophisticatedly. However, it is software because there is no quantitative data or text-level analysis to guide the operationalisation of persuasion constructs for computational measurement. Although the article plays a significant role in contextualising the hypotheses and analytical choices of the present study, it ought to be supplemented by empirical research that directly explores the persuasive consequences and the literary aspects that breed them.

2.2.3 Paper 3: Burtell and Woodside (2023) — Artificial Influence: An Analysis Of AI-Driven Persuasion

Burtell and Woodside (2023) take a broad look at how artificial intelligence (AI) software might revolutionise the field of human persuasion. The authors do not present new empirical evidence but instead outline conceptual avenues that AI may provide to change the balance of persuasive power in society. They point out that most modern AI systems already involve users in ways that can influence behaviour, such as product and video recommendations or the development of continuous dialogic relationships, even though they are not designed with persuasive purposes in mind. The paper argues that, as AI becomes increasingly ingrained in daily communication, its capacity to generate personalised and interesting messages could enable it to conduct individual persuasion at scale, reshaping the creation of beliefs and decision-making in digital environments. This theoretical framing uses examples of chatbots, recommendation engines, and strategic game agents to explain how AI may increase or distort human discourse and the information ecosystem as a whole.

This work has a positive impact on the current study by enabling the prediction of the social and systemic context in which AI-mediated persuasion occurs. In contrast to empirical studies that directly measure persuasive effects, Burtell and Woodside focus on the processes and attendant risks of AI's impact, including the possibility that AI-generated messaging could spread misinformation, shift public discourse, and deprive humans of their autonomy. These terms are relevant to the AI argument-

level and persuasive effectiveness analysis because they highlight that the effects of persuasive messages are tied to linguistic characteristics, scaling, personalisation, and the overall informational context in which they are disseminated.

The main strength of the manuscript is its connection between AI-influenced persuasion and the pressing real-world issues of information integrity and autonomy. It extends beyond this series of single experiments to ask what AI implies as a ubiquitous persuasive force and, therefore, begins the discussion of the long-term consequences of powering the discourse. Such a point of view brings into perspective the rationale behind this thesis: why AI should be studied, why the question of whether it can persuade should be posed, and how and why persuasive effects will vary across contexts and argument types. However, the lack of controlled data or quantified results is a limitation that makes it difficult to apply directly to empirical modelling of persuasion measures. The paper fails to break down specific textual or rhetorical aspects that make an argument more persuasive; therefore, its conclusions should be balanced with empirically validated studies that guide the operationalisation to be conducted in the present study.

2.2.4 Paper 4: Millet et al. (2023) — Defending humankind: Anthropocentric bias in the appreciation of AI art

Millet et al. (2023) examined how people perceive creative outputs, finding that they are evaluated unequally when it can be shown that an AI or a human created the output, a condition they attribute to an anthropocentric mindset that views creativity as a distinctly human attribute. Millet et al. (2023) conducted 4 experiments involving 1,708 participants. In these experiments, repeated artistic pieces, both paintings and music, were shown, but the labels of sources were mixed in such a way that some respondents were told that a given piece was an AI creation, while others were told that it was a human creation. Subjects who assigned authorship to AI consistently rated the artwork as less creative and reported lower awe than those who believed another person had created it. These impacts were acute among those who held firmly to anthropocentric beliefs about creativity. The authors also showed that the mediation by perceived creativity was reduced, indicating that a psychological process based on motivated reasoning was involved in maintaining a human-centred world perception rather than in performing an objective aesthetic judgement.

The study applies to the present project, as it highlights the problem of audience bias arising from source attribution. This variable may affect how readers perceive AI-generated persuasive texts. This bias may reduce the perceived extent of persuasion and confound comparisons between human and AI arguments, as people may systematically undermine outputs attributed to AI. The varied design of the experiment and its massive, already pre-registered sample provide strong evidence for the effect of source labels on perception. As such, the source labels and their impact on perception can offer a conceptual parallel to the study of persuasion, though this study concentrates on artistic evaluation rather than argumentative content.

The strength of the work of Millet et al. (2023) is that it empirically demonstrates the psychological influence of labelling the source of arguments on evaluation processes, thereby predicting possible reactions to arguments generated by AI. Besides, the research situates these impacts within the broader societal discourse on human uniqueness, enriching the understanding of human-AI interaction. However, its drawbacks also deserve mention: it does not directly research persuasion, does not analyse the impact of textual features on attitude change, and instead focuses on cognitive processes that involve aesthetic judgments in the evaluation of a persuasive argument. Although it sheds light on the prejudice against AI-generated outcomes, the paper provides minimal information on the specific textual or rhetorical features that make a message more or less convincing in a data-driven persuasion assignment.

2.3 Key Challenges and Research Gaps

Upon analysing the literature as a unit, several challenges and gaps were identified that dictate the direction of this study. First, in controlled experiments, it is undisputed that AI systems can

manipulate human attitudes, even at scale, a fact supported by studies of large-scale political dialogue in which conversational agents could generate measurable shifts in voter preferences across a variety of countries and issues. These results prove that AI can go well beyond what text-generating tools do and actively manipulate opinions in structured environments where opinions are measured and attitudinal change is recorded before and after. Second, the existing theoretical literature highlights that the interpretation of persuasion messages depends on the actors, the communicative medium, and the underlying message format, which explains why it is important to study persuasive mechanisms independent of the outcome level. This aligns with the need to move beyond outcome measures and investigate how persuasion works.

Psychological effects, such as audience biases, are also a major factor; psychological experiments showing that source attribution leads individuals to underestimate AI outputs by viewing them as machine creations could mitigate persuasion in some instances. Such biases need to be addressed when explaining attitudinal changes, particularly when participants know about AI authorship. Importantly, although the studies verify AI's ability to change attitudes, the number of existing works that indirectly correlate particular textual characteristics and prompt strategies with measurable persuasion outcomes is limited. A conspicuous absence of systematic analysis identifying features of an argument to differentiate changes in ratings without further mapping of the latter remains apparent, which is at the heart of the current approach. The knowledge gained in this section will inform the current study's hypothesis and methodology. The research will examine how persuasiveness varies with textual characteristics, including the density of evidence, rhetorical framing, source cues, and the type of prompts used to elicit argumentative information. The research provides a more detailed insight into the mechanisms underlying differences in persuasiveness ratings, thereby filling a gap in the empirical literature to date and aligning the analysis of the models with theoretical approaches that focus on message quality and audience interpretation.

CHAPTER 3: DATASET

3.1 Dataset Description

The study uses the Persuasion Dataset hosted on Hugging Face at the exact webpage "Anthropic/persuasion". It was created by Anthropic researchers (Esin Durmus, Liane Lovitt, Alex Tamkin, Stuart Ritchie, Jack Clark, and Deep Ganguli) to measure how effectively large language models can change people's minds, as detailed in Anthropic's research note "Measuring the Persuasiveness of Language Models" (9 April 2024). In the original data-collection pipeline, Anthropic curated 28 emerging policy topics and paired supporting and opposing claims, producing 56 opinionated claims. Human-written arguments were produced by randomly assigning three participants to each claim and requesting ~250-word persuasive messages with bonus incentives for compelling writing. Matched model arguments were generated using four prompt styles (Compelling Case, Role-playing Expert, Logical Reasoning, and a Deceptive prompt permitting fabricated facts and sources). Participants then rated agreement before and after reading one argument, yielding pre/post stance scores and an attitude-shift measure; the overall study involved 3,832 unique participants and focused on English-language issues primarily relevant to a U.S. cultural context. This dataset was chosen because it directly links argument text, source, and prompt type to measurable persuasion, enabling controlled comparisons of AI versus human persuasive effectiveness aligned with the research question.

Feature	What it represents	Typical use in analysis
worker_id	Anonymous identifier for the participant who provided both the pre- and post-stance ratings	Group/repeated-measures checks (e.g., controlling for participant-level tendencies)
claim	The policy-relevant statement that the participant is asked to evaluate	Topic/issue controls; claim-level difficulty or contentiousness
argument	The persuasive text presented to the participant (written by a human or generated by a language model)	Text-feature extraction (tone, evidence, length, framing)
source	The author/source of the argument (a model name or "Human")	Comparing human vs AI (and model-to-model) persuasive performance
prompt_type	The prompting strategy used to generate the model argument	Testing which prompting strategies yield larger attitude shifts
rating_initial	Participant's stance/agree–disagree rating before reading the argument	Baseline attitude; covariate for predicting susceptibility to persuasion
rating_final	Participant's stance/agree–disagree rating after reading the argument	Outcome for persuasion; used to compute change (rating_final − rating_initial)

3.2 Exploratory Data Analysis

Exploratory data analysis operationalised **persuasion shift** as *(rating_final − rating_initial)* and visualised its overall distribution, differences by **source** and **prompt_type**, and the relationship between initial and final ratings. The histogram shows most shifts clustered around 0 with a small positive mean (≈0.42), indicating modest average movement after exposure to an argument, while boxplots suggest broadly similar central tendencies across most sources and prompt types, with the control condition centred near zero change. Before analysis, ratings were converted from the dataset's categorical labels (e.g., "1 – Strongly oppose" to "7 – Strongly support") into a numeric 1–7 scale to enable arithmetic shifts and modelling. Argument text was cleaned by lowercasing, removing punctuation, and collapsing whitespace for consistent feature extraction. Missing prompt types (NaN) were replaced with "Unknown," preserving rows for comparison, while rows where ratings could not be mapped (NaN after conversion) were removed to avoid invalid shift values and spurious

correlations. Feature analyses indicate rating_initial is the strongest predictor of rating_final, with text-length features contributing comparatively little.

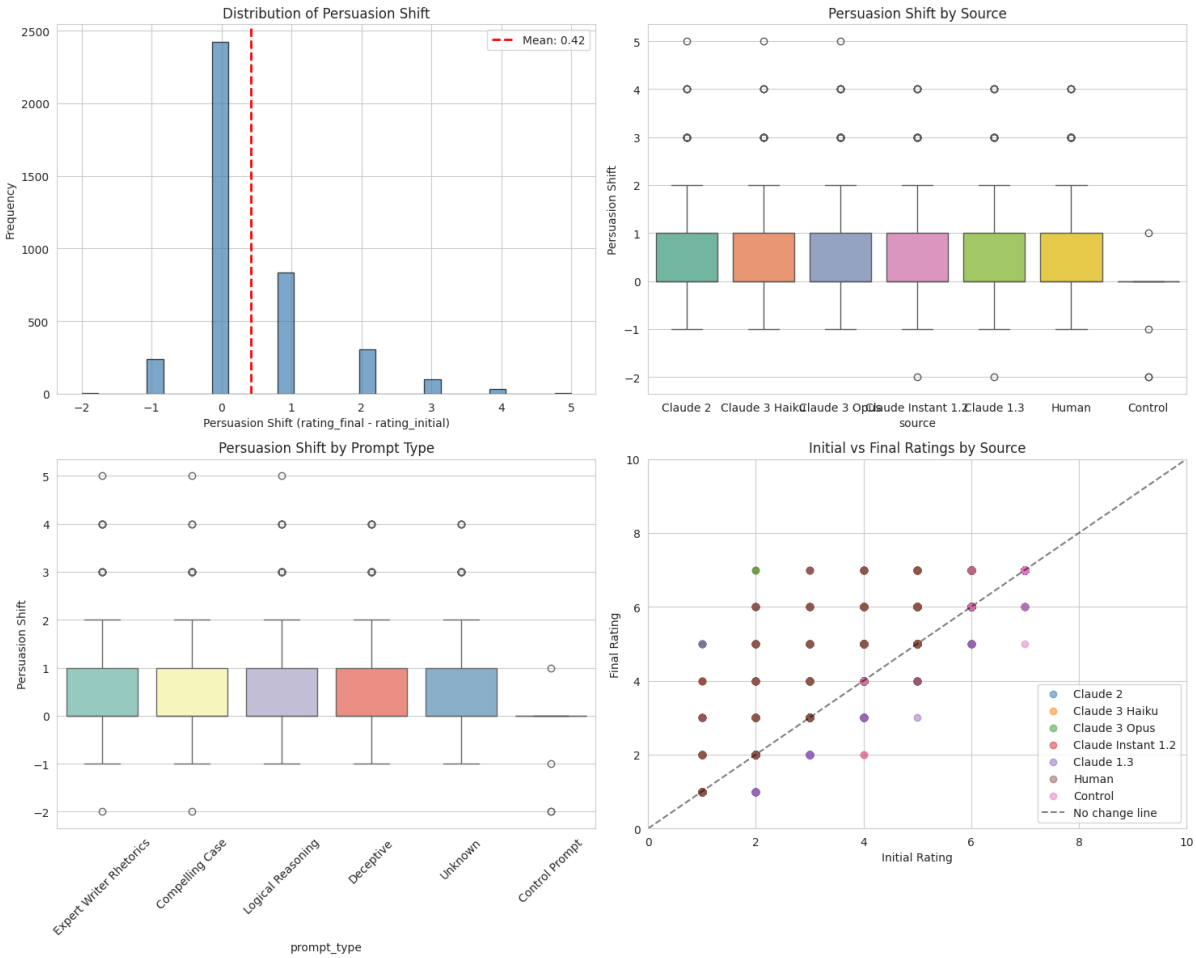


Figure 1: Distribution of persuasion shift, persuasion shift by source, persuasion shift by prompt type, and initial vs final ratings by source

Figure 2: Correlation matrix showing relationships between extracted text features and persuasion metrics.

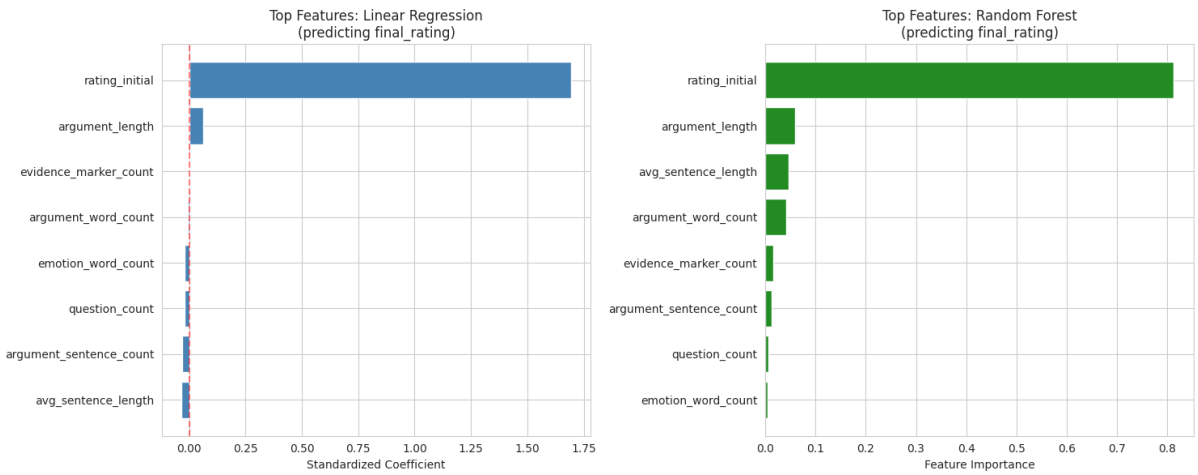


Figure 3: Top predictive features for final rating from linear regression (standardised coefficients) and random forest (feature importance)

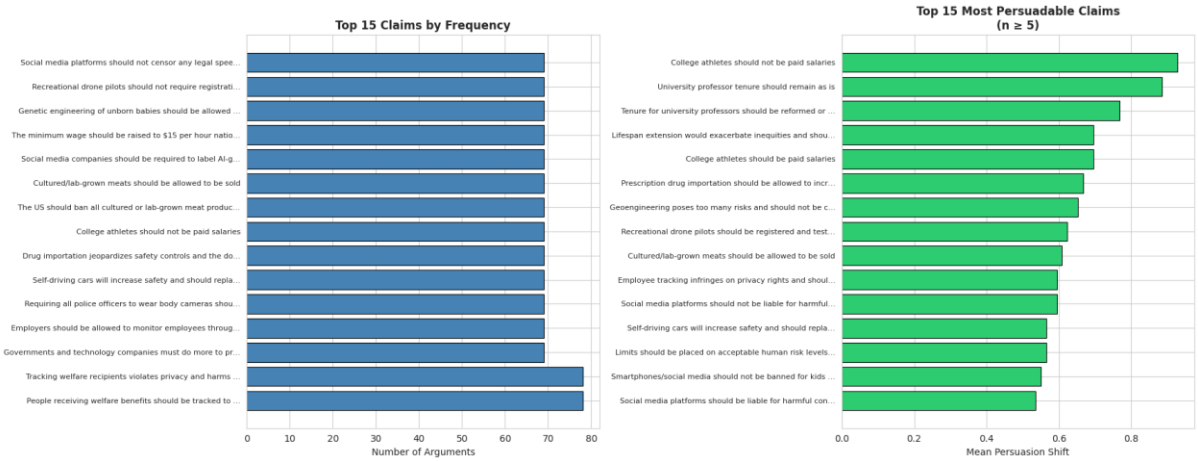


Figure 4: Top 15 Claim/Topic Distribution

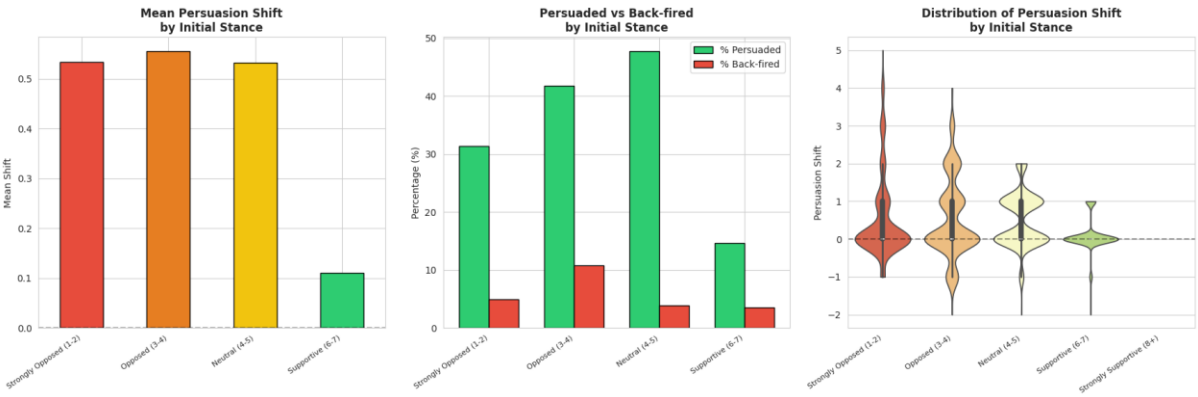


Figure 5:Initial Rating Segment Analysis

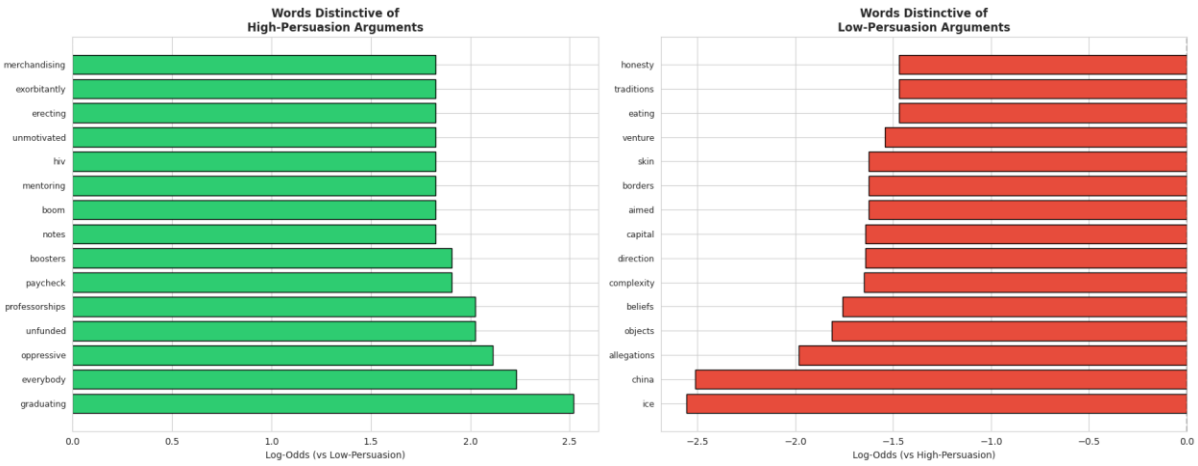
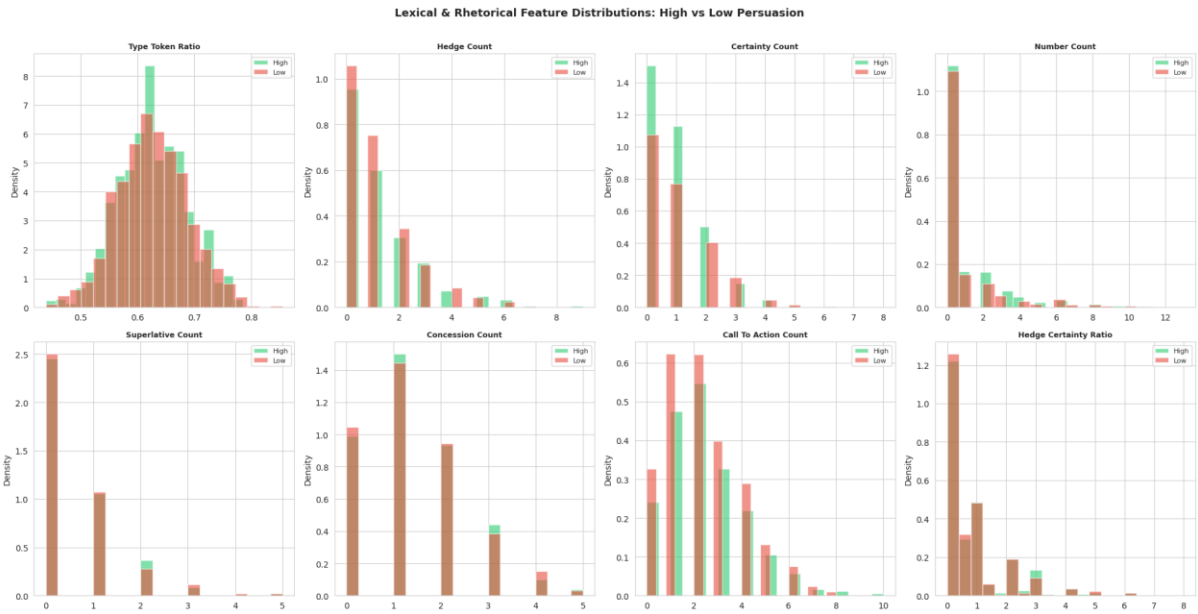


Figure 6:Word Frequency and Bigram Analysis



Lexical and Rhetorical Feature Distributions: High vs Low Persuasion

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